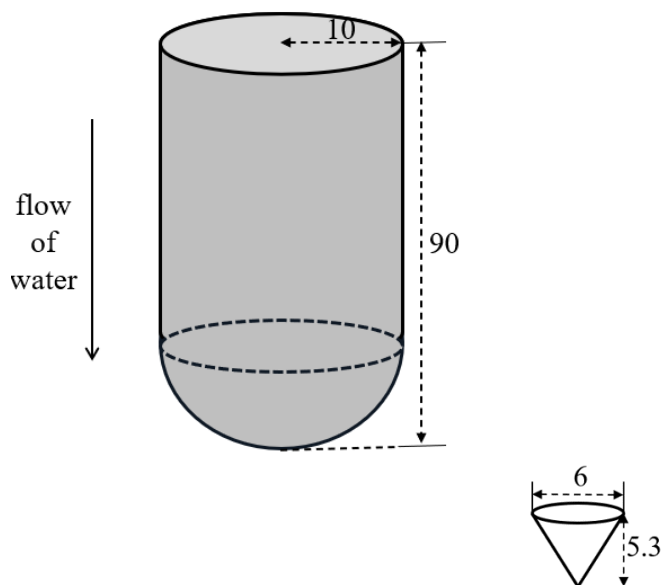


- 6 A water dispenser is in the shape of a cylinder and a hemisphere both of radius 10 cm. The height of the dispenser is 90 cm.

Conical disposable cups of diameter 6 cm and height 5.3 cm are provided to drink the water from the dispenser.



- (a) Water is filled to the brim of the dispenser.

$$8666\frac{2}{3}\pi \text{ cm}^3.$$

Show that the amount of water in the dispenser is
Answer

[2]

- (b) Find the capacity of one conical cup.
 Give your answer to the nearest cm^3 .

Answer _____ cm^3 [2]

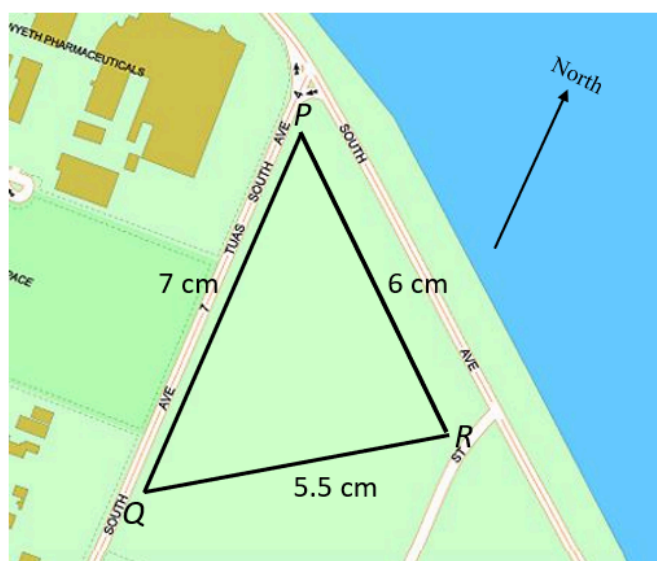
- (c) Find the height of the water remaining in the dispenser after 250 cups of water have been dispensed.

Answer _____ cm [4]

- (d) The conical disposable cups are made of a thin material of negligible thickness. Calculate the cost of 250 conical cups given that the cost of material is 0.003 cents/cm².
Give your answer to the nearest cent.

Answer _____ cents [3]

- 7 The map shown has a scale of 1 : 7500.
An area on the map is formed by a triangle PQR .
 $PQ = 7$ cm, $QR = 5.5$ cm and $RP = 6$ cm.



- (a) Find the actual perimeter, in kilometres, of triangle PQR .

Answer _____ km [2]

- (b) P is due north of Q .
Calculate the bearing of Q from R .

Answer _____ [4]